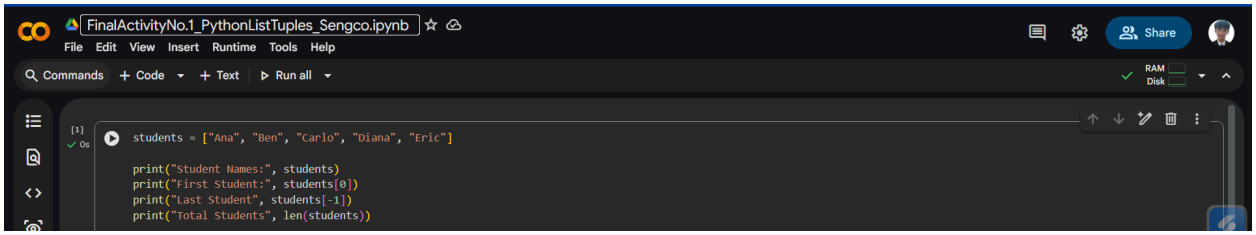
Jose Rizal University
80 Shaw Boulevard, Mandaluyong City
College Division, Computer Science and Engineering

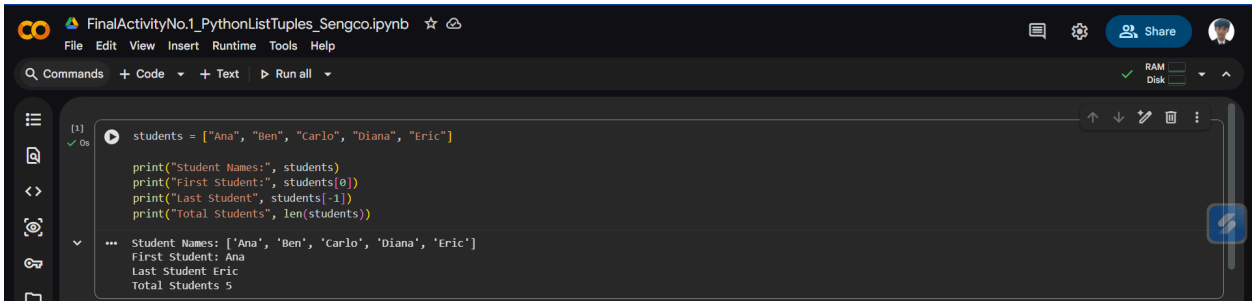
FINAL TERM ACTIVITY NO.1_PythonListTuples

1. Creating and Displaying a List

- Concept: List Creation, Indexing, and Len().



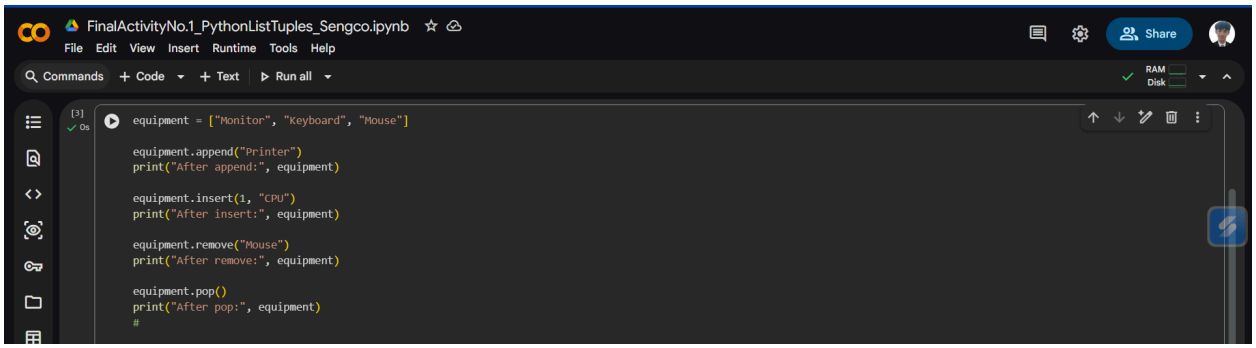
Source Code



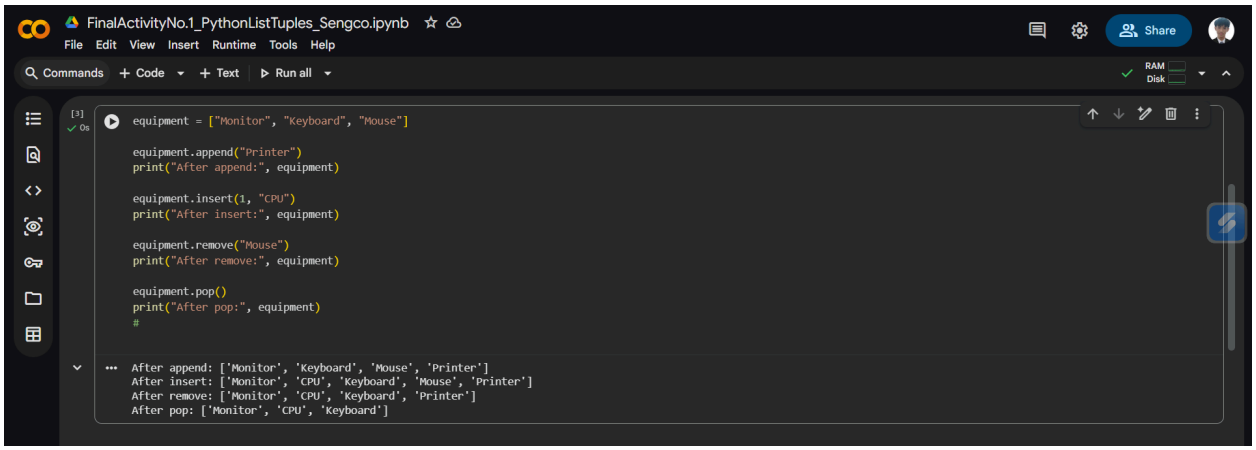
Program Output

2. Adding and Removing List Items

- Concept: Append(), Insert(), Remove(), and Pop()



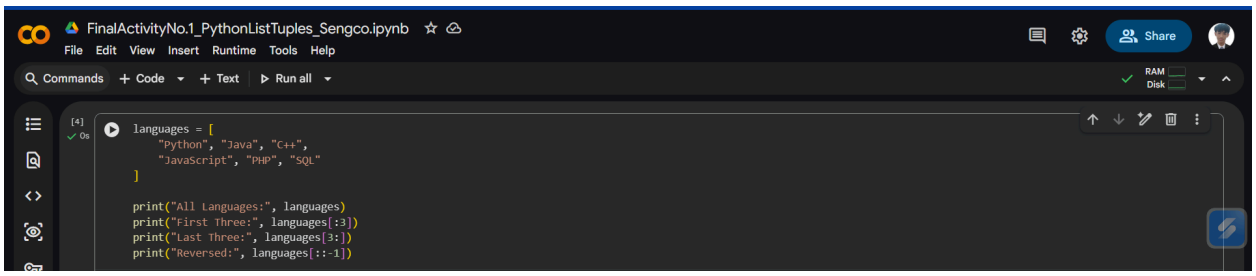
Source Code



Program Output

3. List Slicing

- Concept: Accessing a range of list elements



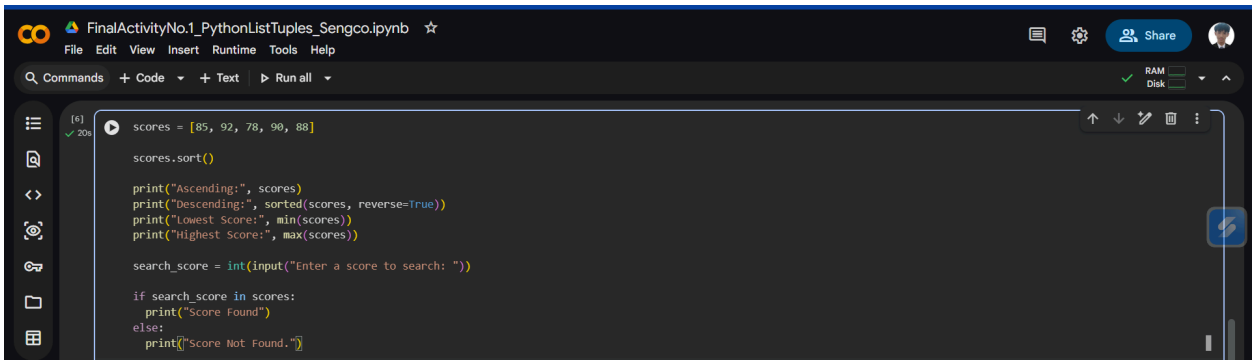
Source Code



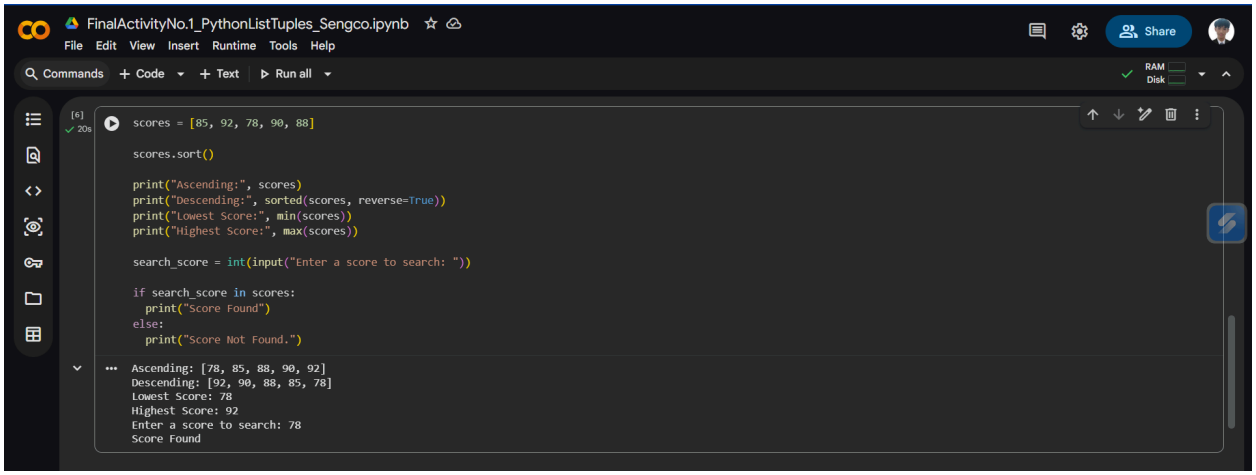
Program Output

4. Sorting and Searching a List

- Concept: Sort(), in, min(), and max().



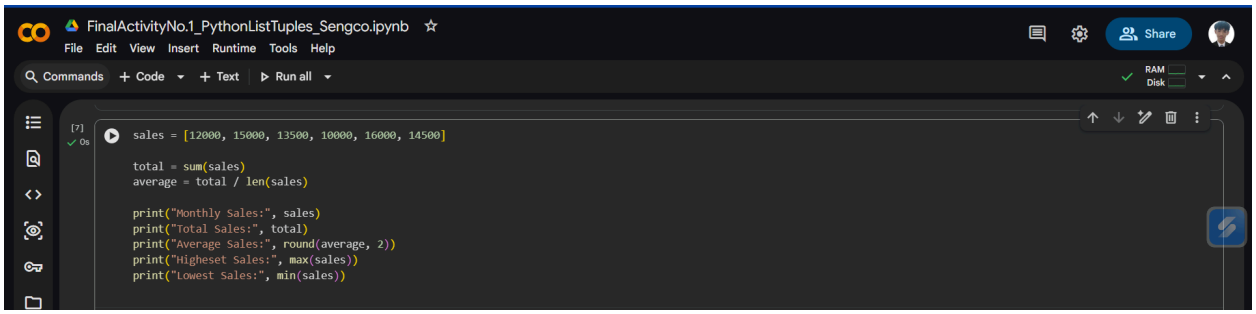
Source Code



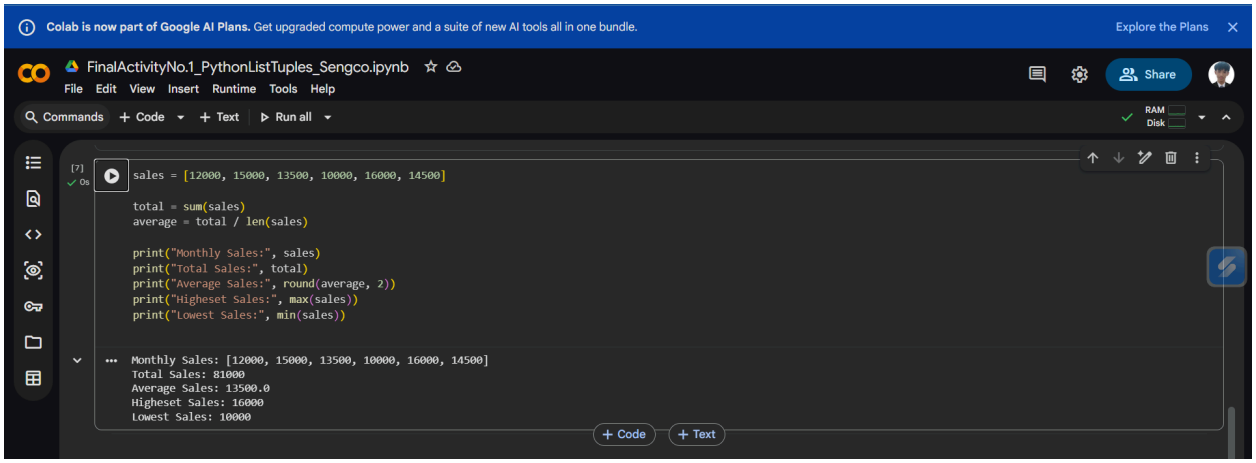
Program Output

5. Computing the Average Using a List

- Concept: sum(), len(), and arithmetic operations.



Source Code



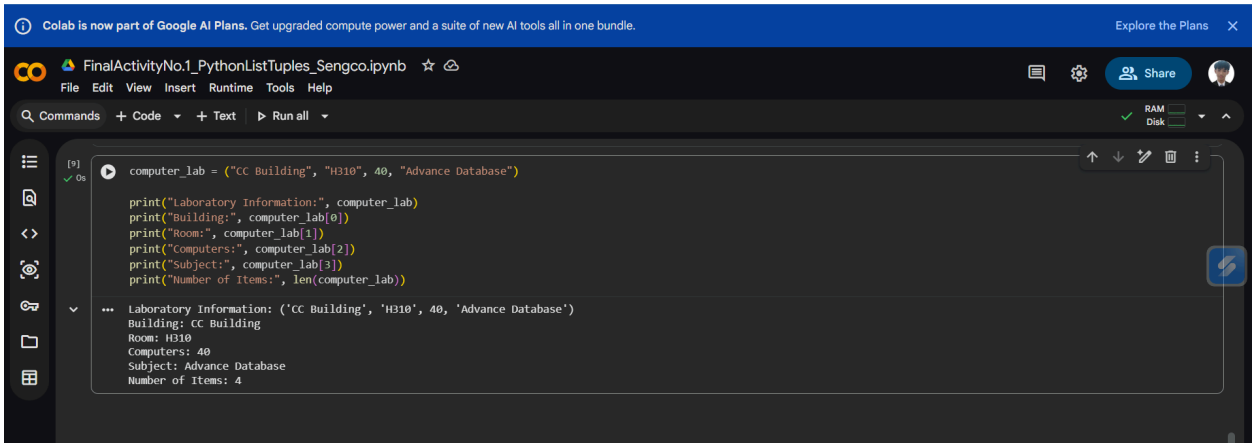
Program Output

6. Creating and Accessing a Tuple

- Concept: Tuple creation, indexing, and immutability.



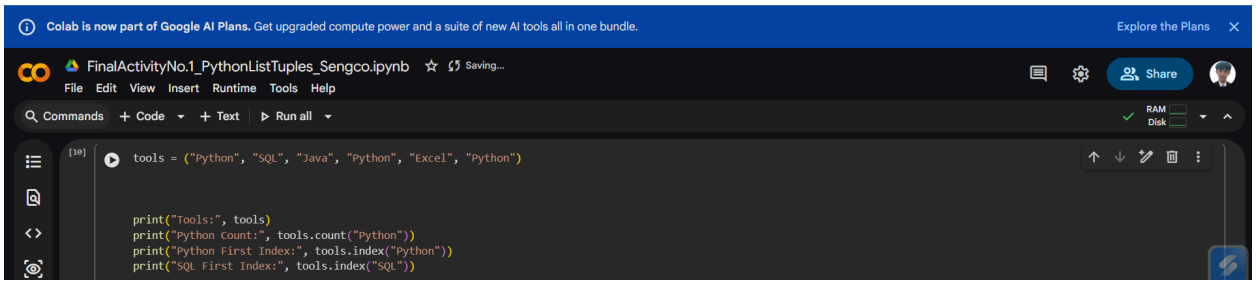
Source Code



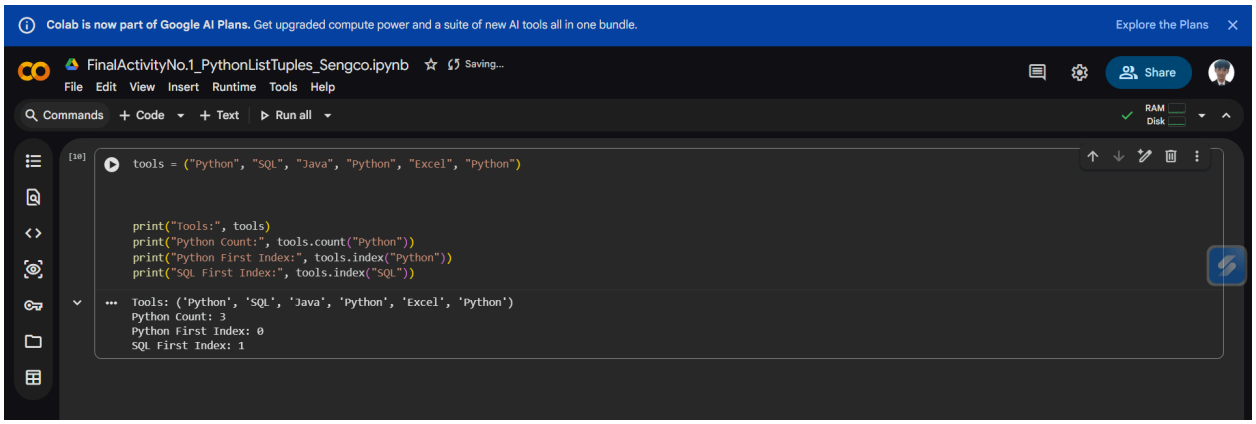
Program Output

7. Tuple Methods: count() and index()

- Concept: Counting occurrences and finding an elements position



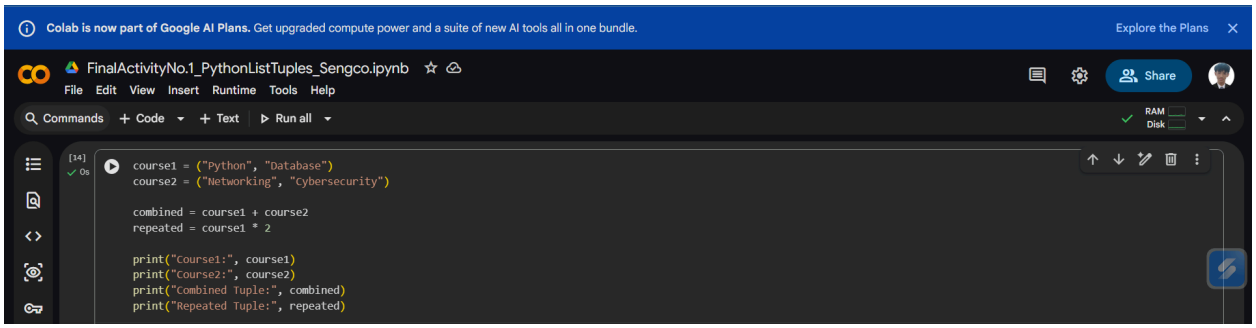
Source Code



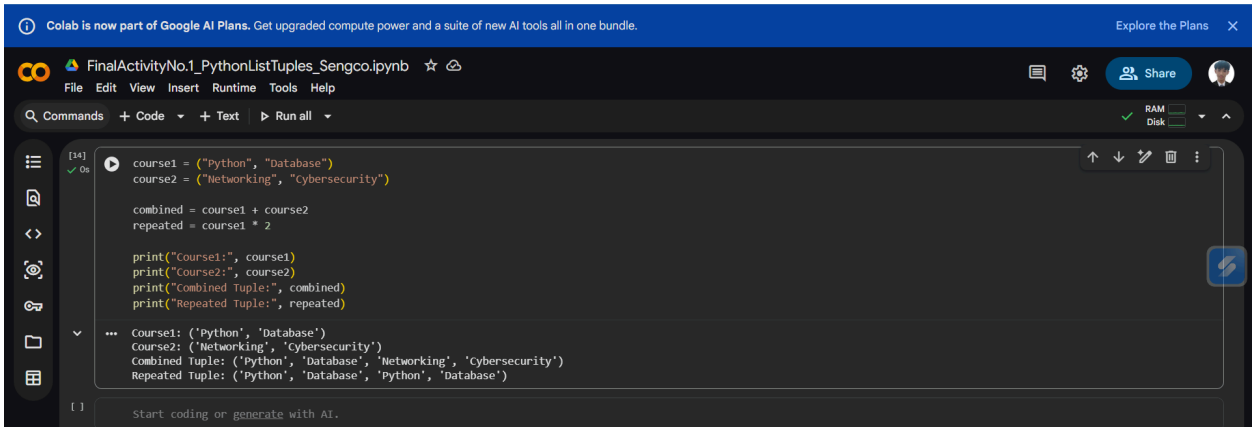
Program Output

8. Combining and Repeating Tuples

- Concept: Tuple Concatenation (+) and Repetition (*)



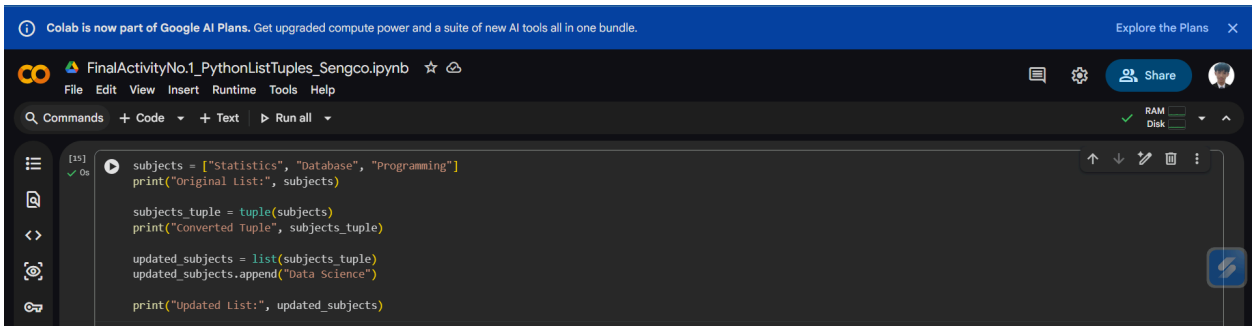
Source Code



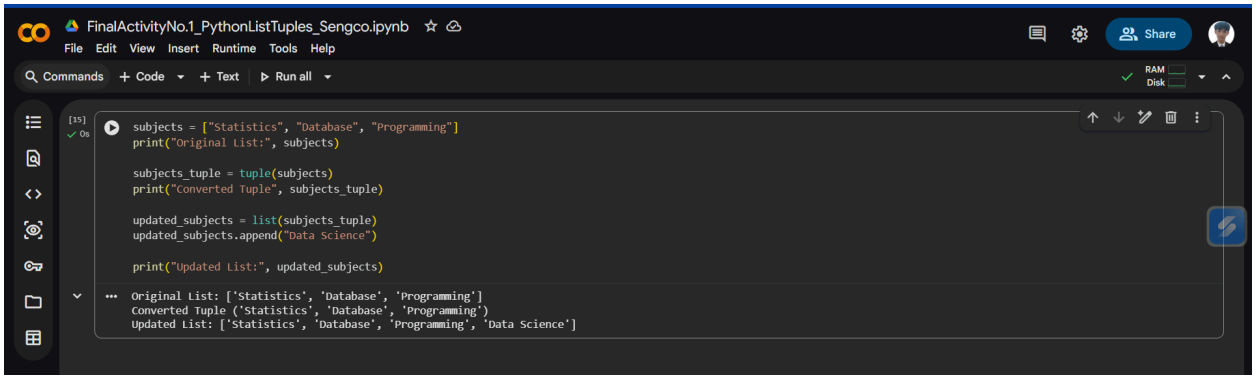
Program Output

9. Converting a List to a Tuple and Back

- Concept: Tuple() and List().



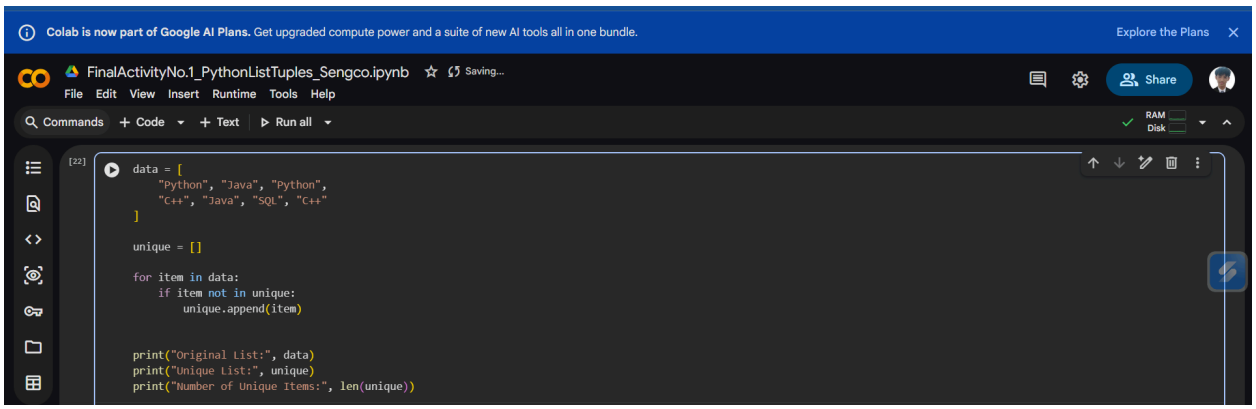
Source Code



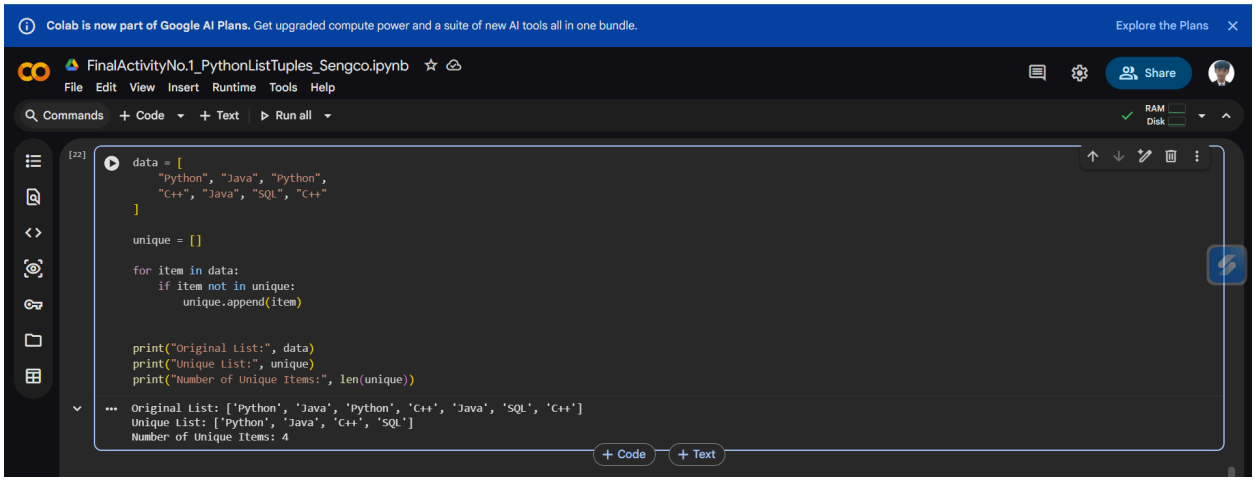
Program Output

10. Removing Duplicate Items from a List

- Concept: Lists, loops, conditional statements, and append()



Source Code



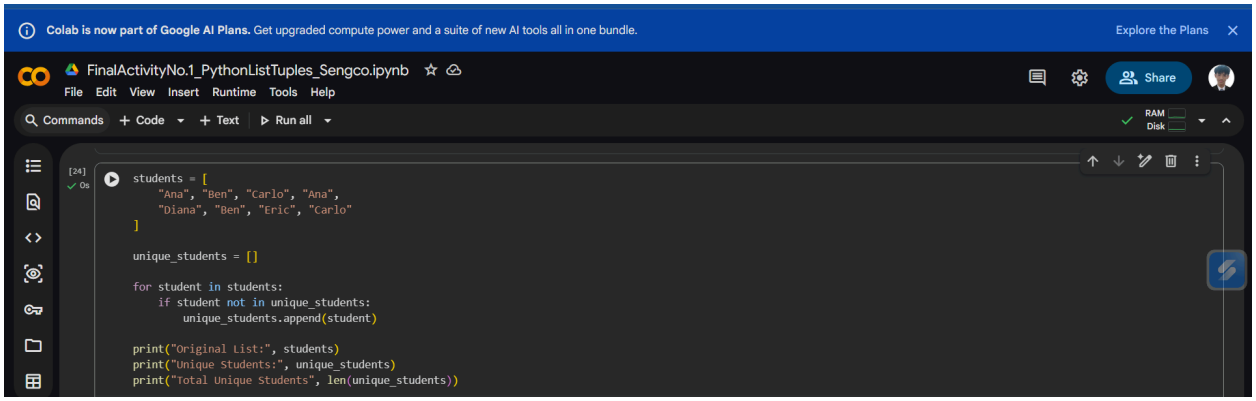
Program Output

10 Difficult-Level Python Sample Codes: List and Tuples

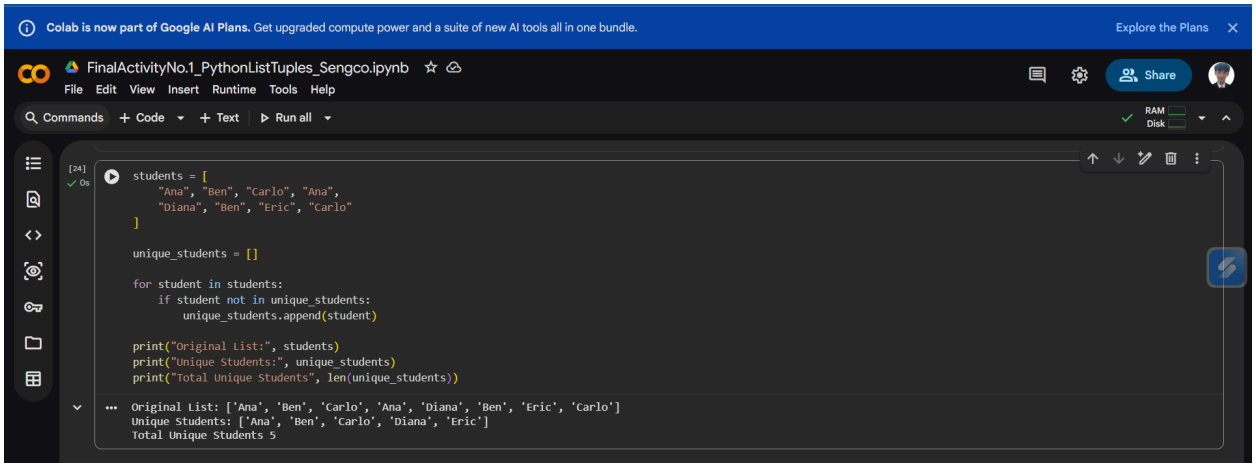
Sample 1: Remove Duplicates While Preserving Order

- Concepts: List, Loops, Membership Testing, Conditional Statements

Problem: Remove duplicates student names without changing the order of their first appearance



Source Code

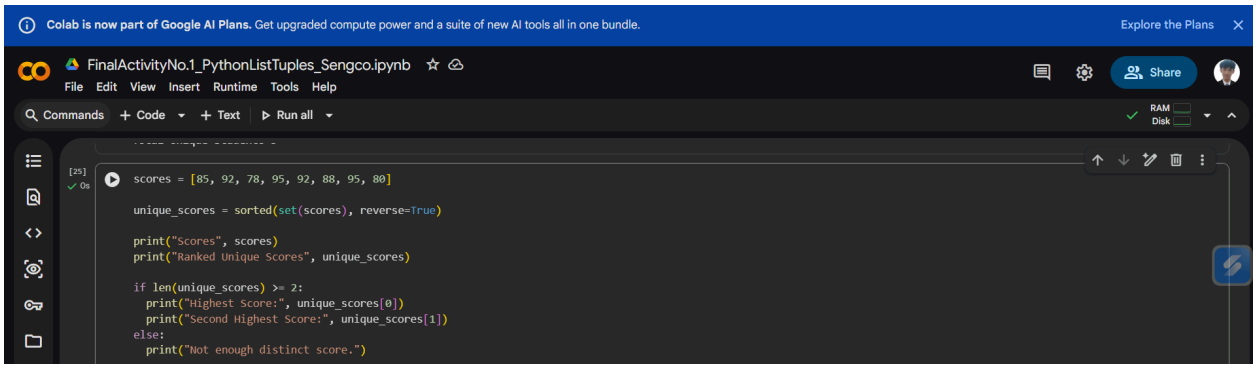


Program Output

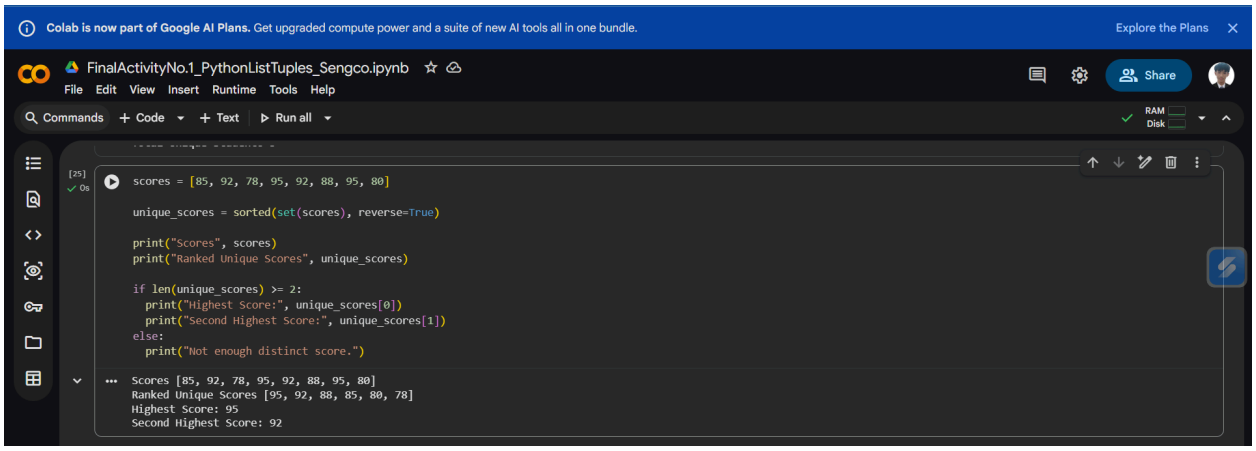
Sample 2: Find the Second-Highest Unique Score

- Concepts: Lists, sets, sorting, duplicate handling.

Problem: Find the second-highest distinct quiz score, even when the highest score appears more than once.



Source Code

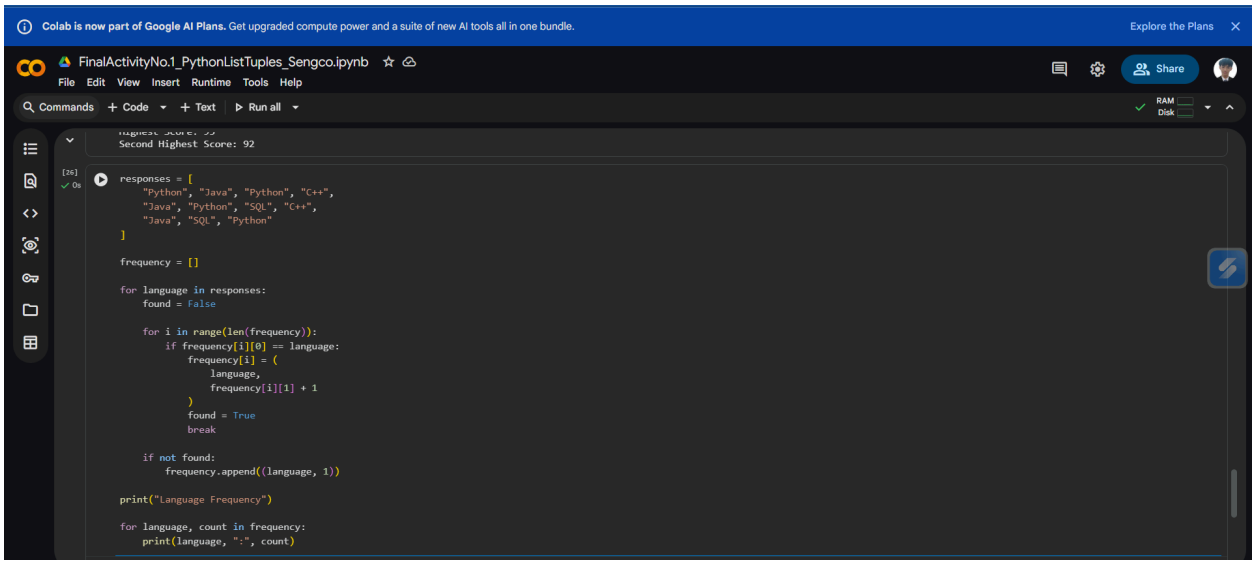


Program Output

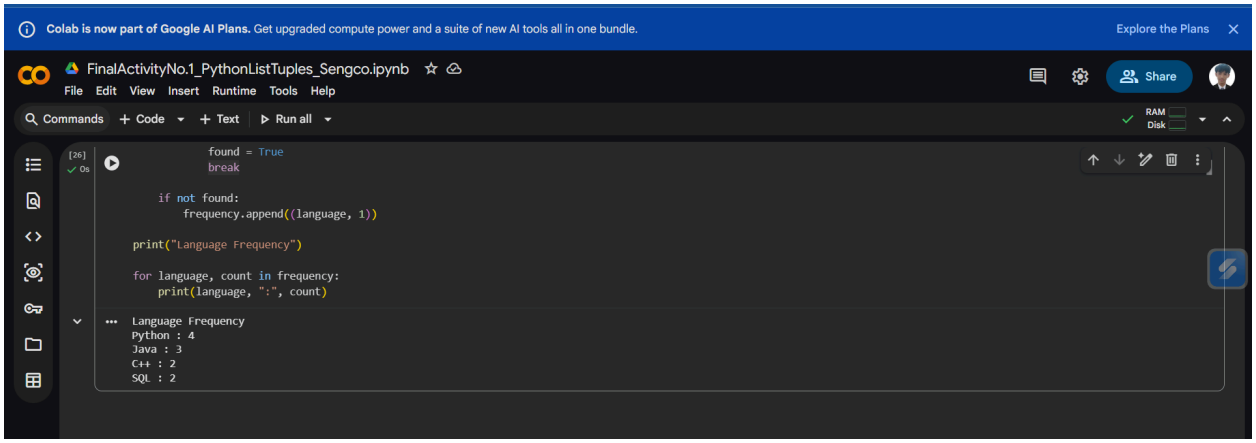
Sample 3: Frequency Distribution Using a List

- Concepts: Lists, tuples, loops, count(), conditional logic.

Problem: Count how many times each distinct programming language appears in a survey response list.



Source Code

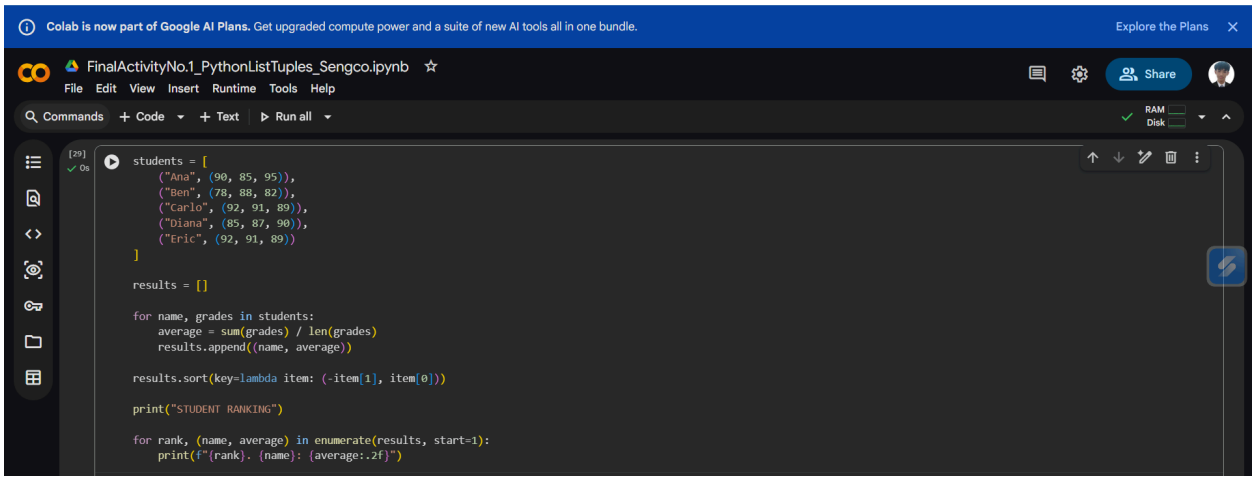


Program Output

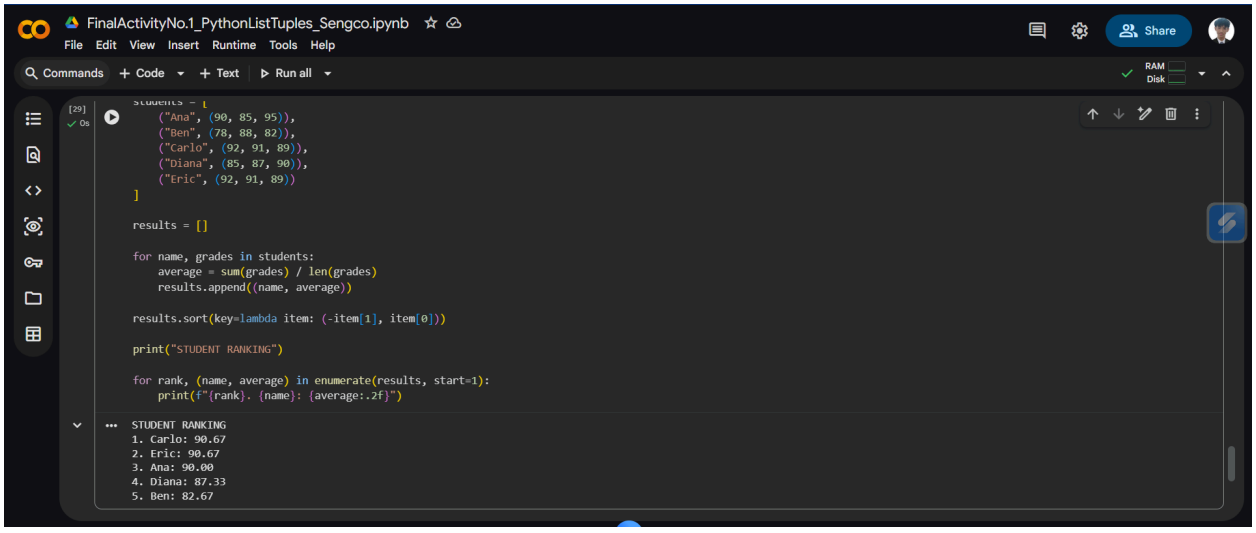
Sample 4: Student Ranking System

- Concepts: Nested tuples, sorting with key, lambda functions.

Problem: Rank students according to their average grades, from highest to lowest.



Source Code

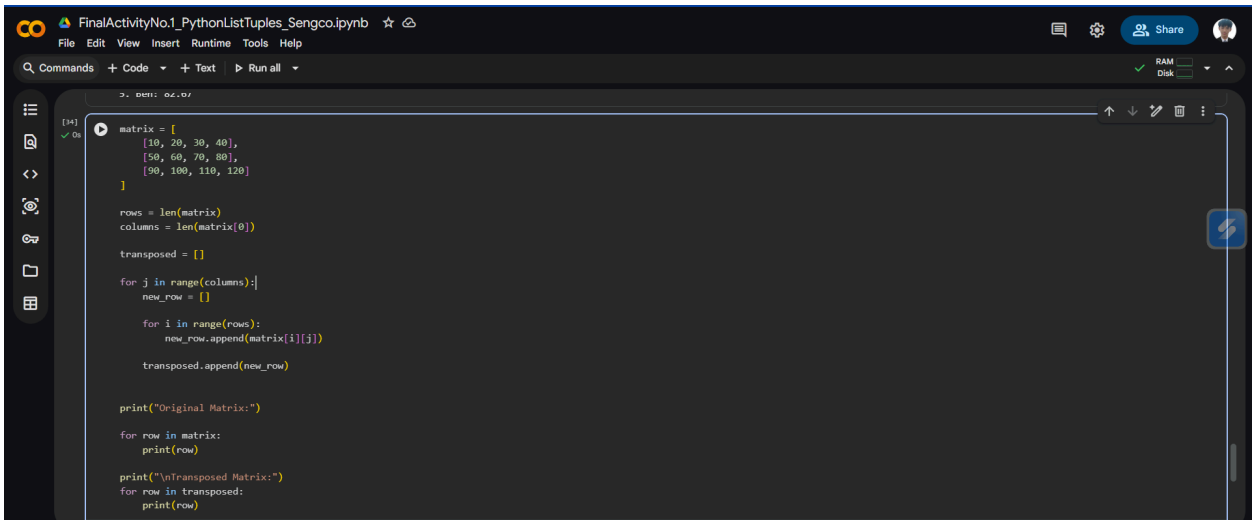


Program Output

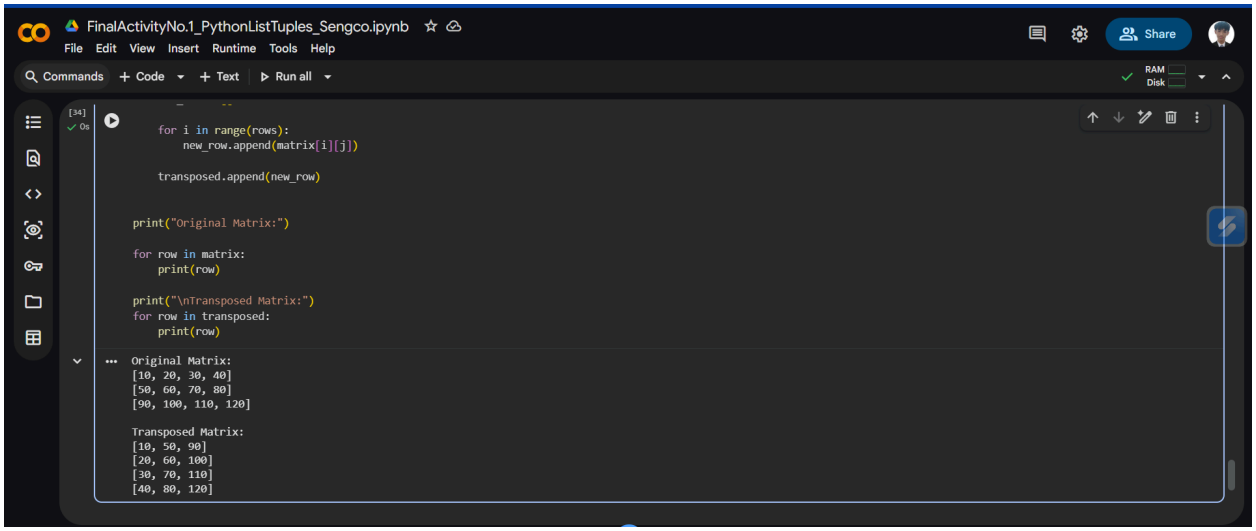
Sample 5: Matrix Transposition Using Nested Lists

- Concepts: Nested lists, indexing, loops, list comprehensions.

Problem: Convert a matrix with three rows and four columns into a matrix with four rows and three columns.



Source Code

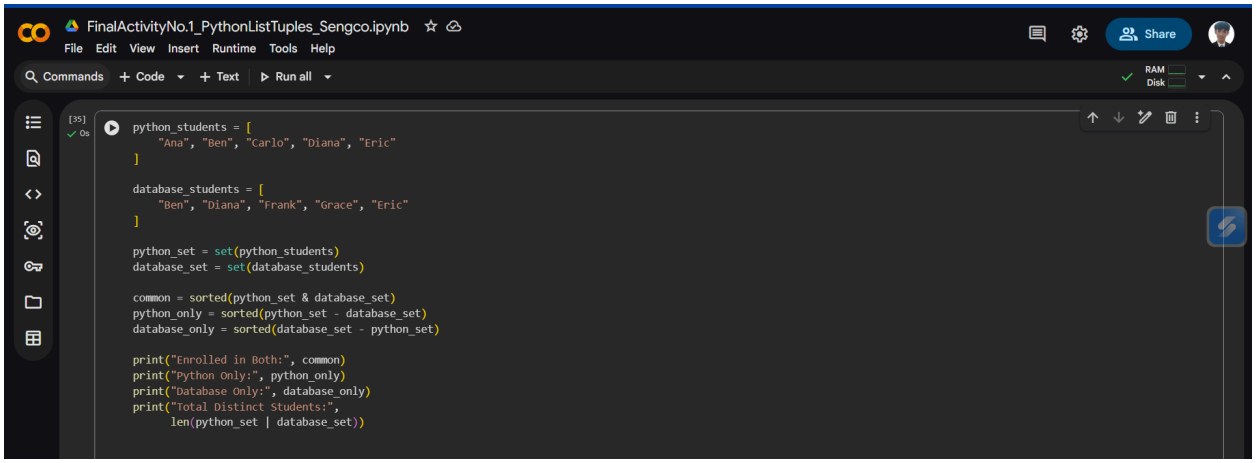


Program Output

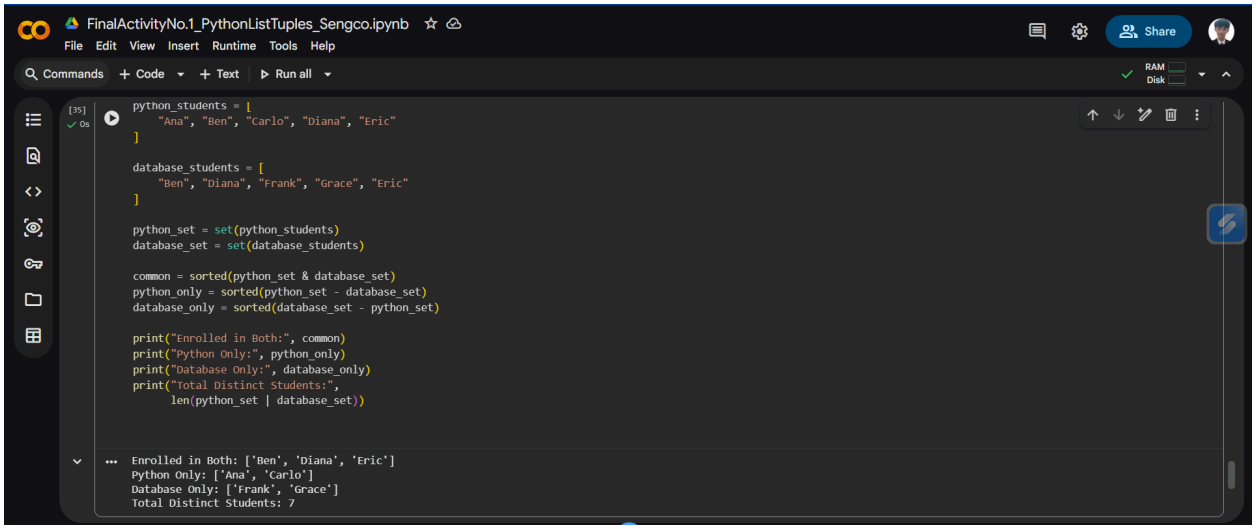
Sample 6: Common and Unique Elements Between Two Lists

- Concepts: Lists, sets, membership testing, set operations.

Problem: Compare two lists of students enrolled in different IT electives.



Source Code

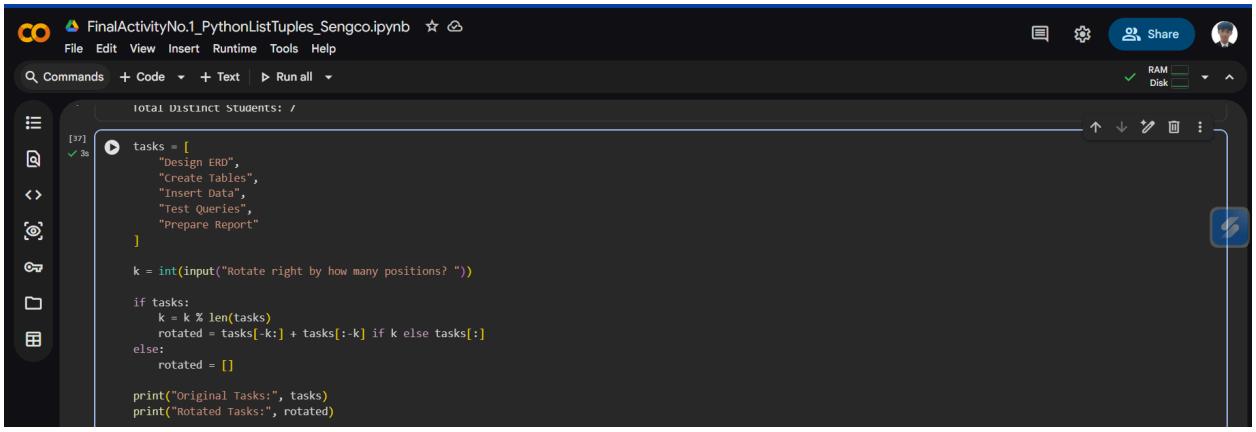


Program Output

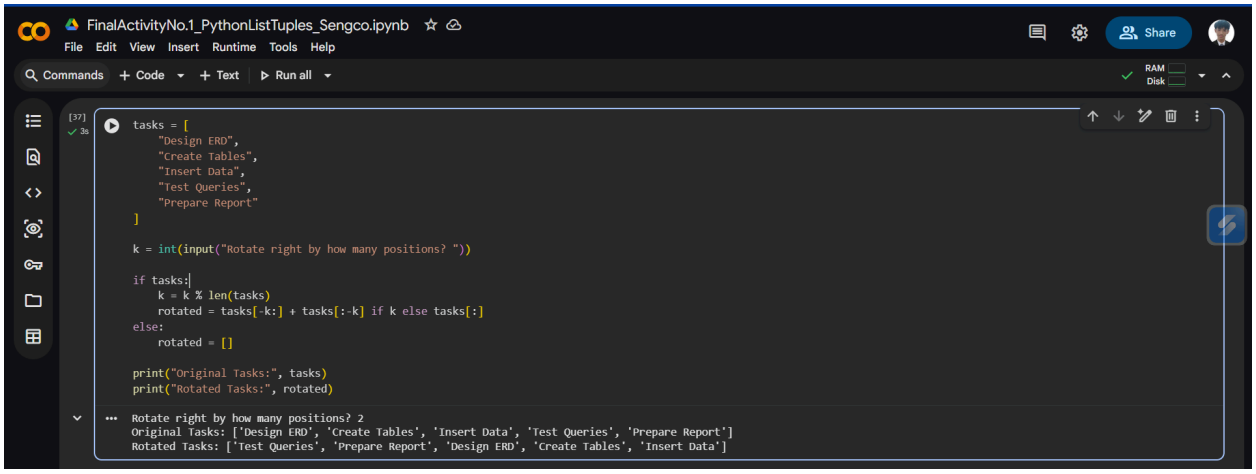
Sample 7: Rotate a List by a User-Defined Number

- Concepts: List slicing, modulo operator, user input.

Problem: Rotate a list of tasks to the right by a number of positions entered by the user.



Source Code

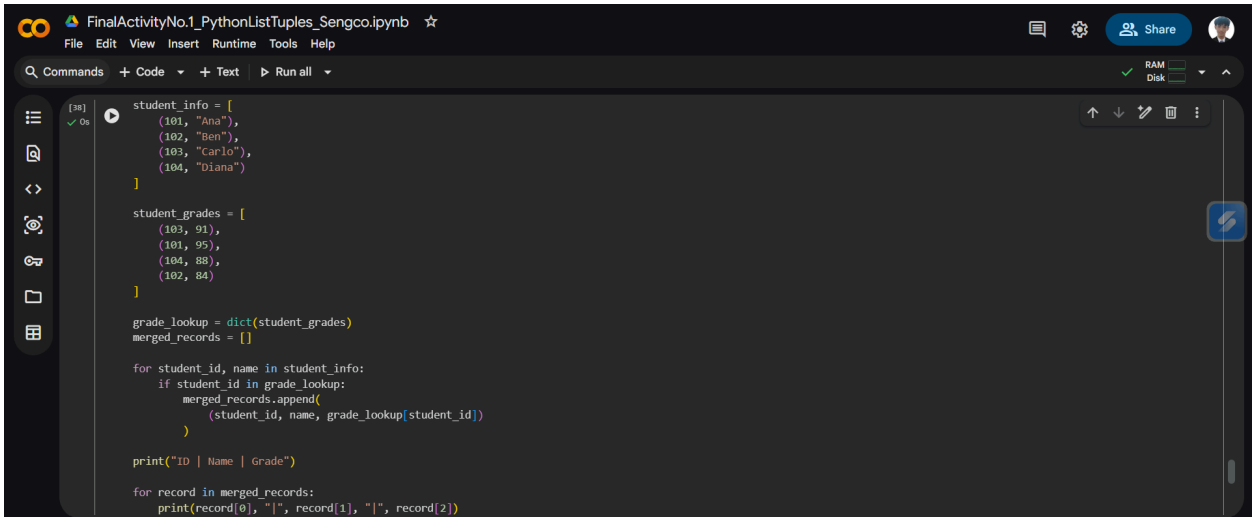


Program Output

Sample 8: Merge Two Lists of Records Using Tuples

- Concepts: Tuples, dictionaries, loops, record matching.

Problem: Merge student information and grades using student ID as the key.



Source Code

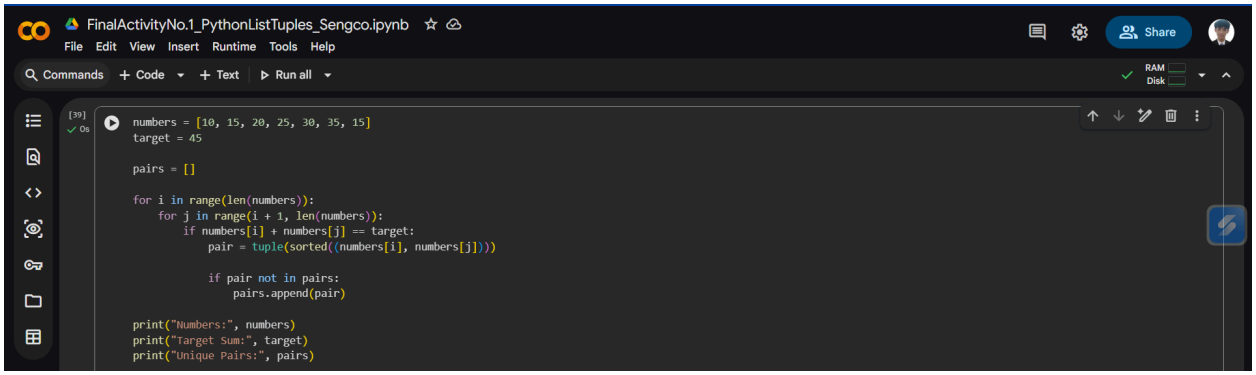


Program Output

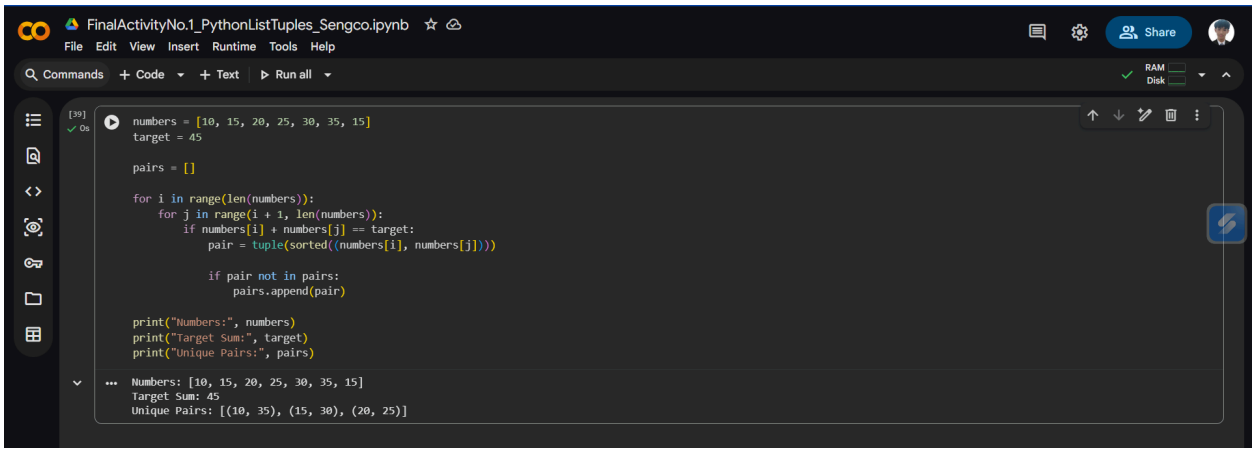
Sample 9: Find Pairs That Equal a Target Sum

- Concepts: Nested loops, tuples, list indexing, conditional statements.

Problem: Find all unique pairs of quiz scores whose sum equals a target value.



Source Code

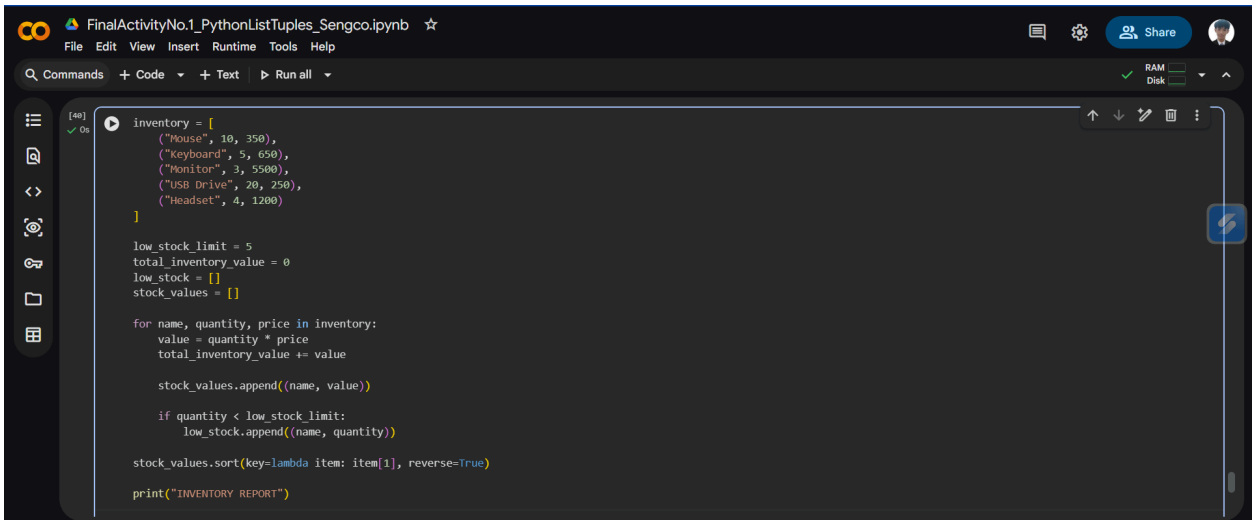


Program Output

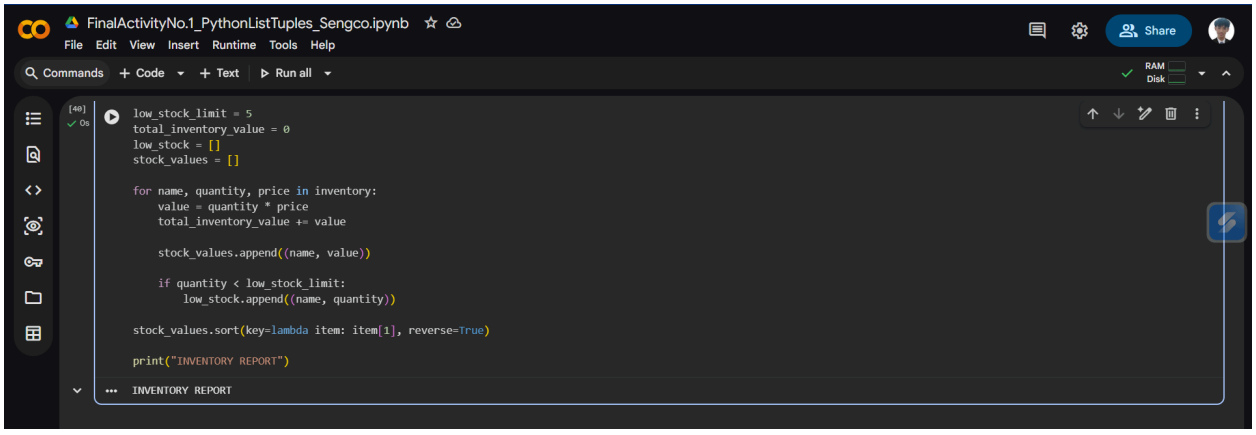
Sample 10: Inventory Management Using Lists and Tuples

- Concepts: Lists of tuples, searching, updating records, filtering, sorting.

Problem: Develop a small inventory program that calculates stock values, identifies low-stock products, and ranks products by total inventory value.



Source Code



Program Output